

KARTHIKEYA SHARMA M

✉ sharmakarhikeya6@gmail.com

🌐 [linkedin.com/in/karthikeyasharma16](https://www.linkedin.com/in/karthikeyasharma16)

🔗 [KarthikeyaSharma16](https://github.com/KarthikeyaSharma16)

Education

Georgia Institute of Technology

08/2023 – 05/2025

Master of Science in Electrical and Computer Engineering

GPA: 3.77/4

SRM Institute of Science and Technology

07/2019 – 05/2023

Bachelor of Technology in Electronics and Communication Engineering

GPA: 9.83/10

Technical Skills

Coding: C/C++, CUDA C, Scripting (Python, Bash, TCL), RTL (Verilog, SystemVerilog), High-Level Synthesis

Libraries: OpenGL, OpenMP, OpenMPI, SFML, PyTorch

Open-source Simulators: ScaleHLS, ScaleSim, TACOS, Chakra, ASTRA-Sim

Tools: Vivado, Vitis & Catapult HLS, Modelsim, Verdi, Icarus Verilog, Libero 12.5, Softconsole, Code Composer Studio

Protocols: I2C, SPI, CAN, UART, AXI (AXI-MM, AXIS, AXI-Lite), QDMA, PCIe

Hardware: Alveo U45N & U55C, Versal, Microsemi SmartFusion2, Pynq-Z2, Raspberry Pi, ARM Mbed, Arduino Uno

Coursework: Advanced Computer Architecture, Hardware-Software Co-Design for ML Systems, Parallel Programming for FPGA, Advanced Programming Techniques, Machine Learning, Digital Systems Testing, Computer Network Security, Computer Communication Networks, ARM Based Embedded System Design, Digital Logic & VLSI Design.

Work Experience

Hardware Engineer

07/2025 – Present

MangoBoost, Inc.

Bellevue, WA

System Architecture & Implementation Team | Hardware Development Group

07/2025 – 12/2025

- Core architect of Data Plane Studio (DPS) for Mango BoostX 400G DPU, enabled scalable reuse of the hardware architecture across multiple use cases (ROCEv2 AI / NVMe-over-RDMA / NTI) etc.
- Owned PCIe and DMA-centric feature development; designed reset-tree architecture and resolved critical hardware functional issues.
- Developed automation scripts to streamline system testing, enhanced CI/CD workflows, prepared package to be delivered for end customers and led comprehensive documentation efforts to support scalable DPU evolution.
- Prepared p

GPU Server Taskforce | Workload Analysis & Performance Optimization

01/2026 – Present

- Application Benchmarking – AI agents (flow of agents, bottleneck analysis), Strix Halo, Guardrails + System optimization
- AI Productization Pipeline + Agent Sandbox
- Identified and characterized application-level GPU server workloads, benchmarking end-to-end performance to expose bottlenecks and guide optimization targets across the software-hardware stack.

Projects

APEX: Collective-Communication | *Python*

04/2024 – 12/2024

- Prototyped **APEX**, an automated tool-chain for analyzing network configurations and collective algorithms for ML algorithms via Design Space Exploration (DSE), revealing 64% of the workload bandwidth and topology dependent.
- Streamlined collective algorithm synthesis by translating TACOS outputs to MSCCLang IR and Chakra Execution Traces, bypassing ASTRA-Sim's system layer reducing engineering effort by 10×.
- Created a configuration file for heterogeneous topology translation and an interactive visualizer for analysis of GPU network collective flows and patterns for congestion and switch modeling.

Cryptonite: Scalable DSE for Cryptographic Workloads | *Python*

01/2024 – 05/2025

- Developed Cryptonite, a tool-chain for generating **hundreds of correct-by-design** HLS designs for FPGA-based accelerators from verified straight-line C implementations of cryptographic primitives.
- Built a DSE engine and QoR estimator for performance analysis, enabling scalable designs with up to 88.88% resource utilization reduction and 54.31% latency improvement.

Federated training and profiling of large-scale ML model | *Python*

03/2024 - 04/2024

- Implemented single and distributed GPU training (2 GPUs) of a GPT-like model on an NVIDIA RTX 6000 cluster using Megatron-LM, utilizing tensor, data, and pipeline parallelism.
- Profiled training with TensorBoard showing tensor parallelism achieving 1.84× speedup over pipeline parallelism.

Performance-Aware Accelerator Design for ML workloads | *Python*

02/2024 - 03/2024

- Conducted DSE using Scale-Sim to optimize systolic array configurations for LeNet CNN, achieving 57% average mapping efficiency for operators while evaluating performance trade-offs across dataflow configurations.
- Developed a mapping strategy (mapping efficiency 84.375%) to mitigate faulty processing elements in a weight-stationary 3x3 systolic array, optimizing GEMM computation latency (54 cycles).

MIPS CPU and CMP memory simulator | *C++*

09/2023 - 12/2023

- Developed a 5-stage (8-pipe) MIPS CPU simulator with Out-of-Order execution, GShare Branch Prediction, and optimized cache hierarchy, reducing CPI by 50% and latency by 10%.
- Engineered a dual-core simulator with L1, L2-shared caches (LRU & random eviction) and DRAM, optimized L2 utilization via SWP & DWP for a 0.85% performance gain.

Publications

- Xu, H., Chen, W., Dixon, J. Z., Maharaj, D. S., **Sharma, K. M.**, Audier, X., Cicerone, M. T. Ultra-high Information-content Chemical Imaging with Broadband Coherent Anti-Stokes Raman and Two-photon Fluorescence Lifetime Microscopy. J. Vis. Exp. (224), e68845, doi:10.3791/68845 (2025). (***Published Oct 7, 2025***)
- **K. S. Maheswaran**, C. Bossut, A. Wana, Q. Zhang and C. Hao, "Cryptonite: Scalable Accelerator Design for Cryptographic Primitives and Algorithms," 2025 IEEE 36th International Conference on Application-specific Systems, Architectures and Processors (ASAP), Vancouver, BC, Canada, 2025, pp. 17-24, doi: 10.1109/ASAP65064.2025.00013. (***Published July 31, 2025***)
- **K. S. M.**, A. Rao, K. Kumar and T. R. Rao, "Terahertz Imaging for Aerospace Applications," 2023 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET), Chennai, India, 2023, pp. 01-05, doi: 10.1109/WiSPNET57748.2023.10134245. (***Published May 30, 2023***)